Project Executable Files

Date	5 july 2026
Team ID	
Project Name	HDI Prediction System using Machine Learning and Flask
Maximum Marks	3 Marks

Project Executable Files

The HDI Insight AI project executable files include the complete source code, trained machine learning model, dataset, templates, static assets, configuration files, and dependency files required to run the application. All project files are organized in a structured manner, enabling the evaluator to execute the Flask-based web application locally without additional modifications. The project predicts the Human Development Index (HDI) using user-provided inputs such as Life Expectancy, Expected Years of Schooling, and Gross National Income. The repository contains all necessary files, including **app.py, train_model.py, model.pkl, requirements.txt, README.md,** and supporting folders (**dataset, templates, static, and model**) to ensure smooth execution and easy maintenance.

Step 1: Submission Checklist

S.No	Item to Submit	Submitted (Yes / No / NA)
1	Complete source code (all files and folders)	Yes
2	README / Setup Guide (instructions to run the project)	Yes
3	requirements.txt / package.json / dependency file	Yes
4	Database schema / seed files (if applicable)	N/A
5	Environment configuration file (.env.example or similar)	No
6	Deployed application URL (if hosted)	No
7	APK / Executable binary (if applicable for mobile/desktop apps)	N/A

8	Dockerfile / Containerization config (if applicable)	No
9	Test files and test results	Yes
10	Demo video or walkthrough (if required)	No

Step 2: File / Folder Structure

HDI_PREDICTION_PROJECT/

```
|
├── 1. Brainstorming & Ideation/
├── 2. Requirement Analysis/
├── 3. Project Design Phase/
├── 4. Project Planning Phase/
├── 5. Project Development Phase/
|   ├── Coding & Solution.pdf
|   ├── Code-Layout, Readability and Reusability.pdf
|   ├── No. of Functional Features Included in the Solution.pdf
|   ├── Performance Testing.pdf
|   ├── Project Executable Files.pdf
|   |
|   └── Source Code/
|       ├── app.py
|       ├── train_model.py
|       ├── requirements.txt
|       |
|       ├── dataset/
|       |   └── HDI.csv
|       |
|       ├── model/
|       |   ├── hdi_model.pkl
|       |   └── imputer.pkl
|       |
|       ├── static/
|       |   ├── style.css
|       |   └── images/
|       |       └── background.jpg
|       |
|       └── templates/
|           └── index.html
|
├── 6. Project Testing/
├── 7. Project Documentation/
└── 8. Project Demonstration/
```

Step 3: Deployment / Access Details

Field	Details
Hosted / Deployed URL	Not Deployed (Local Application)
Login Credentials (Demo)	Not Required
Platform / Hosting Provider	Local Machine (Flask Development Server)
Repository Link	https://github.com/Harishmagodey/HDI-Prediction-System-SmartBridge
Demo Video Link	https://drive.google.com/file/d/1zkjEX5rhW3mV4N74OzeDzQy-pX_WFttF/view?usp=drivesdk

Step 4: Run Instructions

Pre-requisites

- Python 3.10 or above
- pip package manager
- Flask
- Required Python libraries from requirements.txt

Step 1
Clone the repository.
https://github.com/Harishmagodey/HDI-Prediction-System-SmartBridge

Step 2
Install all dependencies.
pip install -r requirements.txt

Step 3
Navigate to the project directory.

Step 4
Run the application.
python app.py

Step 5
Open your web browser.
http://127.0.0.1:5000

Step 6
Enter the required values (Life Expectancy, Expected Years of Schooling, Gross National Income) and click Predict HDI to view the prediction result

Step 5: Known Issues / Limitations

S.No	Known Issue / Limitation	Workaround / Status
1	Application runs only on local server	Limited input parameters
2	Uses a static trained model	Model can be retrained with updated datasets
3	Limited input parameters	Additional socio-economic indicators can be added in future